

GUAVA *and* **BER** **VARIETIES OF** **PAKISTAN**

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PREFACE

Pakistan is blessed with a wonderful blend of agro-climatic conditions suitable for growing all types of tropical, sub-tropical and temperate crops and fruit trees. The major fruit crops grown in different climatic zones in Pakistan include mango, kino and other citrus fruits, date palm, apple, pear and other stone and pome fruits. The fruit trees whether tropical, sub-tropical or temperate are very important economic crops as they are excellent source of all the essential components which help supplement and improve the quality of the human diet. More so the cultivation of fruit crops involves high investments in various phases of production, handling, processing, and marketing. In addition to the nutritional components, many of the species have multi-purpose uses, as they produce non-food products such as fuel, timber, fodder, medicines and industrial products. These activities bring about accelerated economic activity in the agriculture sector which helps alleviate poverty particularly in the rural sector.

Fruiticulture plays a very significant role in crop diversification of farming and agroforestry systems. Their inclusion in farming system reduces the risks against field crop failures. Generally it is said that the farmer's income from fruit orchards is much higher than that from traditional field crop production. However, in Pakistan the potential of tropical, sub-tropical and temperate fruits has not yet been fully realised.

There are a number of reasons for under-realization of the potential; the outstanding ones are non-availability of healthy and true performing nursery trees and absence of regulatory and quality control measures. The purpose of this manual is to provide information on those characteristics

which are necessary to identify the fruit nursery trees in the nurseries and the fields so as to enable application of regulatory and quality control measures, and on those characteristics, which are necessary for raising orchards to maximize yields of best quality, uniform and healthy fruits.

The need was felt since long that the biologically diversified fruit wealth introduced in this part of the world recently or being cultivated since centuries must be described and documented properly. It is appreciated that the Federal Seed Certification and Registration Department has accomplished this remarkable task by producing fruit manuals on mango, citrus, date palm, guava and *ber* fruits.

This manual is on “**Guava and Ber Varieties of Pakistan**” and is quite comprehensive as it covers along with descriptive information on morphological and phenological characteristics of commercial fruit tree varieties, the information about the origin of the variety, maintaining centre and research organizations and leading horticultural characteristics. It is expected that with this basic information at hand, the production system will soon be streamlined to produce and provide sufficient quantities of quality fruit nursery trees to the farming community.

The manuals will further help disseminate very useful information on technical, quality control and production aspects to teachers, students, extension workers, policy makers, fruit tree growers, traders and exporters.

Secretary

Ministry of Food, Agriculture and Livestock

GUAVA VARIETIES

SINDHI

Origin and adaptation: Sindhi, local selection, a chance seedling selection, early season very popular commercial variety of guava in Larkana area, Sindh.

Maintainer: Zaitoon Research Station, Larkana and GPU, PHP, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, well branched, semi spreading to spreading.

Leaf: Foliage medium dense, light green; leaf midrib curvature medium, twisting weak, variegation absent, midrib color of lower side yellowish green, upper surface smooth, lower side pubescence weak, margin undulation weak, length 13-15, width 4-6 and petiole 1-2 cm; tip acute; base obtuse, fragrance weak, flush yellowish green, young leaf anthocyanin medium, pinkish.

Flower: Flowering more than once, mainly spring and autumn, white, fragrance medium, pleasant, style long, stigma high.

Fruit: Skin color (mature fruit) pale yellow, size medium, shape pyriform, fruit long and medium broad, stalk end necked, neck medium long, neck width medium to narrow, stalk cavity medium, apex rounded, fruit length 6-10, breadth 5-7, stalk 1.5 cm, skin thin, surface relief medium rough, longitudinal ridges absent, size of sepals medium, collar around calyx cavity inconspicuous, stalk short, medium thick, fruit muskiness high.

Flesh and seed: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, flesh very thin, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor strong and very pleasant with sweet and wonderful taste.

Seed: Medium seedy, size medium.

Yield: Good bearer.

Leading characters: Sindhi; Early maturing variety, tree spreading, young shoot light green, foliage color light green, fruit long, broad, necked, pale yellow at maturity, surface medium rough, seeds medium, seed size medium, aroma strong, flavor pleasant and sweet.

RIALI

Origin and adaptation: Riali, a popular commercial midseason variety of guava, local selection, a chance seedling selection, mainly grown in Sindh.

Maintainer: Zaitoon Research Station, Larkana and GPU, PHP, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, well branched, branches semi spreading, upright.

Leaf: Foliage medium dense, medium green; leaf midrib curvature medium, twisting weak, variegation absent, midrib color of lower side yellowish green, upper surface smooth, lower side pubescence weak, margin undulation weak, length 14-16, width 4-5 and petiole 1-2 cm; tip acute; base obtuse, fragrance weak, flush yellowish green, young leaf anthocyanin medium.

Flower: Flowering more than once, spring and autumn, size medium, white, fragrance medium, pleasant, style long, stigma high.

Fruit: Skin color (mature fruit) pale yellow, size medium, shape pyriform, fruit long and medium broad, stalk end necked, neck medium long, neck width medium to narrow, stalk cavity medium, apex rounded, fruit length 7-10, breadth 5-7, stalk 1.5 cm, skin thin, surface relief medium rough, longitudinal ridges absent, size of sepals medium, collar around calyx cavity inconspicuous, stalk short, medium thick, fruit muskiness medium.

Flesh and seed: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor medium, pleasant with sweet and wonderful taste.

Seed: Number of seeds few to medium (less than variety Sindhi), usually placed in two circular rows, size medium.

Yield: High.

Leading characters: **Riali** Mid season variety, tree semi spreading to semi drooping, a bit late in maturity, fruit long, broad, necked, pale yellow at maturity, seeds less than variety Sindhi, seeds usually placed in circular rows, aroma medium, flavor pleasant.

RAMZANI

Origin and adaptation: **Ramzani**, a medium commercial, early season guava variety of Larkana area, Sindh, local selection, a chance seedling selection.

Maintainer: Zaitoon Research Station, Larkana and GPU, PHP, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, well branched, branches semi spreading to spreading, semi upright.

Leaf: Foliage medium dense, light green; leaf midrib curvature medium, twisting weak, midrib color of lower side pinkish, upper surface smooth, lower side pubescence weak, margin undulation weak, length 14-16, width 4-6 and petiole 1-2 cm; tip acute; base obtuse, fragrance weak, flush yellowish green, young leaf anthocyanin medium.

Flower: Flowering more than once, spring and autumn, size medium, white, fragrance medium, pleasant, style very long, stigma very high.

Fruit: Size medium, skin color (mature fruit) pale yellow with red spots which increase on over maturity, develops red spots on skin, fruit shape pyriform, long and medium broad, stalk end necked, neck medium long, neck width medium to narrow, stalk cavity medium, apex rounded, fruit length 6-9, breadth 4-6, stalk 1.5 cm, skin thin, surface relief medium rough, longitudinal ridges absent, size of sepals medium, calyx cavity shallow, collar around, stalk short, medium thick, fruit muskiness medium.

Flesh and flavor: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor medium, pleasant with sweet and wonderful taste.

Seed: Medium seedy (more than variety Sindhi) arranged in semi circular rows, size medium.

Yield: Good bearer.

Leading characters: **Ramzani**, Early maturing variety, tree semi spreading to spreading, fruit long, medium broad, necked, pale yellow at maturity, develops red spots on the skin which darkens on over maturity, slightly more number of seeds than Sindhi, usually arranged in semi-circular irregular rows, aroma medium, flavor pleasant.

THADARAMI

Origin and adaptation: Thadarami, a medium popular, medium late maturing commercial variety of guava of Larkana area; local selection, a chance seedling selection.

Maintainer: Zaitoon Research Station, Larkana and GPU, PHP, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, well branched, branches semi spreading to spreading.

Leaf: Foliage medium dense, light green; leaf midrib curvature medium, twisting weak, midrib color of lower side yellowish, upper surface smooth, lower side pubescence weak, margin undulation weak, length 12-14, width 4-6 and petiole 1-2 cm; tip acute; base obtuse, fragrance weak, flush yellowish green, young leaf anthocyanin medium.

Flower: Flowering more than once, spring and autumn, size medium, white, fragrance weak, pleasant, style very long, stigma high.

Fruit: Size medium, skin color (mature fruit) pale yellow, fruit shape pyriform, long and medium broad, stalk end necked, neck medium long, neck width medium to narrow, stalk cavity medium, apex rounded, fruit length 8-13, breadth 5-8, stalk 1-4 cm, skin thin, surface relief medium rough, bumpy, longitudinal ridges present, size of sepals medium, calyx cavity shallow, collar around, stalk short, medium thick, fruit muskiness medium.

Flesh and seed: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor medium, pleasant with sweet and wonderful taste.

Seed: Medium seedy, size medium.

Yield: Good bearer.

Leading characters: Thadarami, Medium late maturing variety, tree semi spreading to spreading, fruit large, long, broad, medium long necked, pale yellow at maturity, longitudinal ridges prominent, surface medium rough, bumpy also called *Karila amrood*, flesh white, medium seedy, aroma medium, flavor rich, sweet, pleasant.

GULABI

Origin and adaptation: Gulabi, a medium commercial variety of guava (*Psidium guajava*); local selection, a chance seedling selection, medium early variety of Larkana area, Sindh.

Maintainer: Zaitoon Research Station, Larkana and GPU, PHP, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, well branched, branches upright to semi spreading, flush yellowish green.

Leaf: Foliage medium dense, dark green; leaf midrib curvature weak, twisting weak, midrib color of lower side light green, upper surface smooth, lower side pubescence weak, margin undulation weak, length 12-15, width 4-6 and petiole 1-2 cm; tip acute; base obtuse, fragrance weak, flush light green, young leaf margin anthocyanin medium.

Flower: Flowering more than once, spring and autumn, size medium, white, fragrance medium, style medium long, stigma height medium.

Fruit: Size medium, skin color (mature fruit) pale yellow, fruit shape roundish, stalk cavity medium, apex rounded, fruit length 6-9, breadth 4-5, stalk 1.5 cm, skin thin, surface relief medium rough, collar around calyx cavity inconspicuous, stalk short, medium thick, fruit muskiness medium to strong.

Flesh and seed: Flesh color red, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor medium, pleasant with sweet and wonderful taste.

Seed: Medium seedy, size medium.

Yield: High.

Leading characters: **Gulabi** Medium maturing variety, tree upright to semi spreading, foliage dark green as to others light to medium green, fruit roundish, pale yellow at maturity, stalk short, surface medium rough, collar inconspicuous, flesh red, texture fine, very sweet, medium seedy, seeds white, medium, aroma medium to strong, flavor pleasant.

SEEDLESS GUAVA

Origin and adaptation: Seedless, *Bedana*, local selection, a chance seedling selection from Dargai area, midseason low scale variety.

Maintainer: GPU, Sherkhana, ARI, Tarnab, NWFP.

Tree: Small tree, medium height, vigorous growing, stem medium thin, well branched, branches semi spreading upright.

Leaf: Foliage medium dense, medium green; leaf broad, midrib curvature medium, twisting absent, midrib color of lower side medium green, upper surface smooth, lower side pubescence weak, margin undulation weak, length 11-14, width 5-8 and petiole 1-2 cm; tip acute; base obtuse, fragrance weak, flush light green, young leaf margin anthocyanin medium.

Flower: Flowering more than once, spring and autumn, size medium, white, fragrance medium, style long, stigma high.

Fruit: Size medium, skin color (mature fruit) pale yellow, fruit shape pyriform, roundish, medium broad, stalk end neck very weak to absent, stalk cavity medium, apex rounded, fruit length 6-9, breadth 5-8, stalk 1.5 cm, skin thin, surface relief medium rough, bumpy, longitudinal ridges present, size of sepals medium, calyx cavity shallow, collar around, stalk short, medium thick, fruit muskiness medium.

Flesh and seed: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness medium, texture fine, medium firm, flavor fair, aroma weak, pleasant.

Seed: Seedless

Yield: Medium bearer.

Leading characters: Seedless Guava; Midseason low scale variety, tree upright, semi spreading, fruit roundish broad, neck very weak to absent, pale yellow at maturity, longitudinal ridges prominent, surface bumpy seedless, medium sweet, aroma weak, pleasant, flavor fair, may be a suitable variety for salad purpose.

PG 001

Origin and adaptation: PG 001, Introduction from Bangla Desh, a seedling selection, midseason variety of guava.

Maintainer: GPU, Sherkhana, ARI, Tarnab, NWFP.

Tree: Small bushy tree, medium height, vigorous growing, stem medium thin, well branched, semi spreading to spreading.

Leaf: Foliage medium dense, medium green; leaf midrib curvature medium, variegation absent, midrib color of lower side light green, upper surface smooth, lower side pubescence very weak, margin undulation weak, length 13-17, width 4-7 and petiole 3-4 cm; tip acute; base obtuse, fragrance absent, flush light green, young leaf anthocyanin medium, color pinkish.

Flower: Flowering more than once, mostly spring and autumn, color white, style long, stigma high, fragrance medium, pleasant.

Fruit: Skin color (mature fruit) greenish white, size medium, shape roundish, stalk cavity medium, apex rounded, fruit length 6-8, breadth 5-8, stalk 1.5 cm, skin thin, surface relief medium rough, longitudinal ridges absent, size of sepals medium, collar around calyx cavity inconspicuous, stalk short, medium thick, fruit muskiness high.

Flesh and seed: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness weak, puffiness weak, juiciness medium, acidity low, sweetness medium, texture fine, medium firm, flavor medium, pleasant.

Seed: Medium seedy, size medium.

Yield: High.

Leading characters: PG 001; Midseason variety, introduced from Bangla Desh, tree small bushy, size medium, semi spreading to spreading, young shoot light green, foliage color medium green, flower fragrance medium, pleasant, style long, stigma high, fruit size medium, roundish, broad, neck absent, color greenish white at maturity, aroma medium, flesh color white, sweetness medium, flavor medium good, pleasant, Medium seedy, seed size medium.

PG 005

Origin and adaptation: PG 005, Introduction from Bangla Desh, a seedling selection, midseason variety of guava.

Maintainer: GPU, Sherkhana, ARI, Tarnab, NWFP.

Tree: Small tree, bushy, height medium, vigorous, stem medium thin, well branched, branches semi spreading to semi drooping.

Leaf: Foliage medium dense, light green; leaf midrib curvature weak, twisting weak, variegation absent, midrib color of lower side yellowish green, upper surface smooth, lower side pubescence very weak, margin undulation absent, length 13-15, width 6-8 and petiole 2-5 cm; tip acute; base obtuse, fragrance absent, flush light green, young leaf margin anthocyanin medium.

Flower: Flowering more than once, mostly spring and autumn, color white, style long, stigma high, fragrance weak to medium, pleasant.

Fruit: Skin color (mature fruit) greenish white, size medium, shape roundish, stalk cavity medium, apex rounded, fruit length 6-9, breadth 5-8, stalk 1-2 cm, skin thin, surface relief medium rough, longitudinal ridges absent, size of sepals medium, calyx cavity shallow, collar around, stalk short, medium thick, fruit muskiness medium.

Flesh and seed: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness weak, puffiness weak, juiciness medium, acidity low, sweetness medium, texture fine, medium firm, flavor medium, pleasant.

Seed: Medium seedy, size medium.

Yield: High.

Leading characters: PG 005; Midseason variety, introduced from Bangla Desh, tree small bushy, size medium, semi spreading to spreading, young shoot light green, foliage color medium green, flower fragrance weak to medium, pleasant, style long, stigma high, fruit size medium, roundish, broad, neck absent, color greenish white at maturity, aroma medium, flesh color white, sweetness medium, flavor medium good, pleasant, Medium seedy, seed size medium.

PG 10

Origin and adaptation: PG 10, Introduction from Bangla Desh, a seedling selection, midseason variety of guava.

Maintainer: GPU, Sherkhana, ARI, Tarnab, NWFP.

Tree: Small tree, bushy, height medium, vigorous, stem medium thin, well branched, branches semi spreading to semi drooping.

Leaf: Foliage medium dense, dark green; leaf midrib curvature medium, twisting weak, midrib color of lower side light green, upper surface smooth, lower side pubescence weak, margin undulation absent, length 12-14, width 4-6 and petiole 1-2 cm; tip acute; base obtuse, fragrance absent, flush yellowish green, young leaf margin anthocyanin medium.

Flower: Flowering more than once, spring and autumn, size medium, white, fragrance medium, pleasant, style long, stigma high.

Fruit: Skin color (mature fruit) greenish white, size medium, shape roundish, stalk cavity medium, apex rounded, fruit length 6-8, breadth 5-7, stalk 1.5 cm, skin thin, surface relief medium rough, longitudinal ridges absent, size of sepals medium, calyx cavity shallow, collar around, stalk short, medium thick, fruit muskiness medium.

Flesh and seed: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness weak, puffiness weak, juiciness medium, acidity low, sweetness medium, texture fine, medium firm, flavor medium, pleasant.

Seed: Medium seedy, size medium.

Yield: High.

Leading characters: PG 10; Midseason variety, introduced from Bangla Desh, tree small bushy, size medium, semi spreading to spreading, young shoot light green, foliage color dark green, flower fragrance medium, pleasant, style long, stigma high, fruit size medium, roundish, broad, neck absent, color greenish white at maturity, aroma medium, flesh color white, sweetness medium, flavor medium good, pleasant, Medium seedy, seed size medium.

PG 13

Origin and adaptation: PG 13, Introduction from Bangla Desh, a seedling selection, midseason variety of guava.

Maintainer: GPU, Sherkhana, ARI, Tarnab, NWFP.

Tree: Small tree, bushy, height short, medium vigorous, medium upright, stem medium thin, well branched, branches semi spreading.

Leaf: Foliage dense, medium green; leaf midrib curvature absent, twisting absent, midrib color of lower side light green, upper surface smooth, lower side pubescence weak, margin undulation absent, length 12-14, width 4-6 and petiole 1-2 cm; tip acute; base obtuse, fragrance weak, flush yellowish green, young leaf margin anthocyanin medium.

Flower: Flowering more than once, spring and autumn, size medium, white, fragrance medium, style taller than PG 005, stigma higher than PG 005.

Fruit: Skin color (mature fruit) greenish white, size medium, shape roundish, stalk cavity medium, apex rounded, fruit length 5-8, breadth 5-7, stalk 1.5 cm, skin thin, surface relief medium rough, longitudinal ridges absent, size of sepals medium, calyx cavity medium, collar around, stalk short, medium thick, fruit muskiness high.

Flesh and seed: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness weak, puffiness weak, juiciness medium, acidity low, sweetness medium, texture fine, medium firm, flavor medium, pleasant.

Seed: Medium seedy, size medium.

Yield: High.

Leading characters: PG 13; Midseason variety, introduced from Bangla Desh, tree small bushy, size short, semi spreading, young shoot light green, foliage color medium green, flower fragrance medium, pleasant, style very long, stigma height longer than PG 005, fruit size medium, roundish, broad, neck absent, color greenish white at maturity, aroma medium, flesh color white, sweetness medium, flavor medium good, pleasant, Medium seedy, seed size medium.

KAZI

Origin and adaptation: Kazi, Introduction from Bangla Desh, a seedling selection, midseason variety of guava.

Maintainer: GPU, Sherkhana, ARI, Tarnab, NWFP.

Tree: Small tree, bushy, height medium, vigorous, stem medium thin, well branched, branches semi spreading, young branch anthocyanin strong, pinkish.

Leaf: Foliage medium lax, light green; leaf midrib curvature medium, twisting absent, midrib color of lower side yellowish, upper surface smooth, lower side pubescence weak, margin undulation weak, length 8-10, width 5-6 and petiole 1-2 cm; tip acute; base obtuse, veins sunken, fragrance absent, flush light green, young leaf margin anthocyanin strong.

Flower: Flowering more than once, mainly spring and autumn, size medium, white, fragrance weak, pleasant, style short, stigma height same as staminal column.

Fruit: Skin color (mature fruit) greenish white, size medium, shape roundish, stalk cavity medium, apex rounded, fruit length 6-8, breadth 5-7, stalk 1.5 cm, skin thin, surface relief medium rough, longitudinal ridges absent, size of sepals medium, calyx cavity shallow, collar around calyx cavity inconspicuous, stalk short, medium thick, fruit muskiness high.

Flesh and seed: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness weak, puffiness weak, juiciness medium, acidity low, sweetness medium, texture fine, medium firm, flavor medium, pleasant.

Seed: Medium seedy, size medium.

Yield: High.

Leading characters: Kazi; Introduced from Bangla Desh, tree small bushy, young shoot anthocyanin strong, foliage light green, leaves small, flower style short, stigma height same as staminal column, fruit size medium, roundish, light greenish white at maturity, aroma medium, flesh color white, sweetness medium, flavor medium good, pleasant, Medium seedy, seed size medium.

BER VARIETIES

BAHAWALPUR SUFAID

Origin and adaptation: Bahawalpur Sufaid, a good commercial variety of *Ber*, (*Zizyphus mauritania*); a chance seedling selection, medium early variety of Bahawalpur zone with long picking season.

Maintainer: Horticultural Research Station, Bahawalpur and GPU, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, shoot thin, well branched, branch attitude spreading droopy, medium spiny, spine size medium, young shoot color brown.

Leaf: Foliage medium dense, medium green, leaf size medium, elongate, midrib curvature medium, twisting weak, midrib color of lower side green, upper surface smooth, lower side pubescence strong, margin undulation very weak, length 8.5, width 5.5 and petiole 1.0 cm; tip broad; base broad, flush yellowish green, young leaf anthocyanin weak.

Flower: Flowering more than once, spring and autumn, size medium.

Fruit: Size medium large, skin color (mature fruit) white, ripened reddish, stalk cavity large, deep, fruit shape roundish, apex somewhat narrow, fruit length 4.0, breadth 4.0, stalk 1.0 cm, skin thin, surface relief medium smooth, longitudinal streaks present, prominence medium, color reddish, fruit muskiness medium.

Flesh: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor very good, pleasant sweet.

Seed: Stone size medium small and rough.

Yield: High.

Leading characters: Bahawalpur Sufaid plant small tree of medium size, well branched, spreading, droopy, medium spiny, spines medium, fruit size medium bold, roundish, white at maturity, golden yellow at ripening turning chocolate brown on over ripening, longitudinal reddish streaks prominence medium, surface smooth, flesh white, texture fine, uniform, stone medium small, aroma medium, flavor pleasant very sweet.

BS 1

Origin and adaptation: Bahawalpur Selection 1 (BS 1), a new variety of *Ber*, (*Zizyphus mauritania*); grown on commercial scale in Bahawalpur area, local selection, a chance seedling selection, mid season variety.

Maintainer: HRS, Bahawalpur.

Tree: Height medium, umbrella shape, vigorous growing, stem medium thin, well branched, shoot thin, branches long, semi spreading to droopy, low spiny, spine size medium bold, young shoot color medium brown.

Leaf: Foliage medium dense, medium green; leaf size medium, elongate, midrib curvature low, twisting medium, midrib color of lower side green, upper surface smooth, lower side pubescence very strong, margin undulation weak, length 5.0, width 3.0 and petiole 2.0 cm; tip roundish; base obtuse, flush yellowish green, young leaf anthocyanin medium.

Flower: Flowering more than once, spring and autumn, size medium.

Fruit: Size medium bold, resembles Bahawalpur white, skin color (mature fruit) pale green, ripened brownish yellow, stalk cavity small, fruit shape oblong, apex somewhat narrowed, fruit length 4.0, breadth 4.0, stalk 1.0 cm, skin thin, surface relief smooth, longitudinal streaks present, medium prominent, color reddish, fruit muskiness medium.

Flesh: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor good, pleasant medium sweet.

Seed: Stone size medium small.

Yield: High.

Leading characters: **BS 1** (Bahawalpur Selection 1). Plant small tree of medium size, well branched, semi spreading to droopy, fruit resembles Bahawalpur Sufaid but a brownish yellow, less sweet and high yielding. longitudinal reddish streak weak, surface smooth, flesh white, texture fine, uniform, medium firm, stone medium small, rough, acidity low, aroma medium, flavor pleasant, medium sweet.

GILI

Origin and adaptation: Gili, a medium commercial variety of *Ber*, (*Zizyphus mauritania*); local selection, a chance seedling selection, medium early variety of Sindh.

Maintainer: Jujuba Research Station, HRI, Mirpurkhas and GPU, DI Khan.

Tree: Small tree, medium umbrella height, vigorous growing, stem medium thin, well branched, shoot thin, spreading, semi droopy, medium spiny, spine size medium bold, young shoot color brown.

Leaf: Foliage medium dense, light green; leaf size medium, elongate, midrib curvature medium, twisting weak, midrib color of lower side green, upper surface smooth, lower side pubescence very strong, margin undulation weak, length 6.0, width 3.0 and petiole 2.0 cm; tip roundish; base obtuse, flush yellowish green, young leaf anthocyanin weak.

Flower: Flowering more than once, spring and autumn, size medium.

Fruit: Size medium bold, skin color (mature fruit) pale green, ripened pale yellow turning to chocolate brown on over maturity, stalk cavity small, fruit shape fusiform, long and medium broad, base and apex narrowed, fruit length 6.0, breadth 3.4, stalk 1.0 cm, skin thin, surface relief medium smooth, longitudinal streaks absent, medium prominent, color reddish, fruit muskiness weak.

Flesh: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor medium, pleasant with sweet and wonderful.

Seed: Stone size medium small.

Yield: High.

Leading characters: Gili plant small tree of medium size, well branched, semi spreading to droopy, fruit medium bold, long, broad in the middle, pale green at maturity, red at ripening, longitudinal reddish streaks medium prominent, surface smooth, flesh white, texture fine, uniform, medium firm, stone size medium small, aroma medium, flavor pleasant.

GOAR

Origin and adaptation: Goar, a medium commercial variety of *Ber*, (*Zizyphus mauritania*); a chance seedling selection, medium early variety of Bahawalpur zone.

Maintainer: Horticultural Research Station, Bahawalpur and GPU, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, shoot thin, well branched, branch attitude spreading droopy, medium spiny, spine size medium, young shoot color brown.

Leaf: Foliage medium dense, light green, leaf size medium, midrib curvature medium, twisting weak, midrib color of lower side green, upper surface smooth, lower side pubescence strong, margin undulation very weak, length 8.5, width 5.5 and petiole 1.0 cm; tip broad; base broad, flush yellowish green, young leaf anthocyanin weak.

Flower: Flowering more than once, spring and autumn, size medium.

Fruit: Size small, skin color (mature fruit) pale green, shiny, ripened reddish brown, stalk cavity narrow, fruit shape round, base and apex round, fruit length 4.0, breadth 3.2, stalk 1.0 cm, skin thin, surface relief medium smooth, longitudinal streaks present, prominence medium, color reddish, fruit muskiness weak.

Flesh: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness low, acidity low, sweetness low, texture fine, medium firm, flavor medium, pleasant with acid tinge.

Seed: Stone size medium small, smooth.

Yield: High.

Leading characters: Goar plant small tree of medium size, well branched, spreading droopy, medium spiny, spines medium, fruit size medium, round oval, pale green at maturity, reddish brown at ripening, longitudinal reddish streaks prominence medium, surface smooth, flesh white, texture fine, uniform, stone medium small, aroma medium, flavor pleasant.

GOLA LEMAI

Origin and adaptation: Gola Lemai, a medium commercial variety of *Ber*, (*Zizyphus mauritania*); a chance seedling selection, medium late variety of Sindh.

Maintainer: Jujuba Research Station, HRI, Mirpurkhas and GPU, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, shoot thin, well branched, branch attitude spreading semi droopy, medium spiny, spine size medium, young shoot color brown.

Leaf: Foliage medium dense, medium green; leaf size medium, midrib curvature medium, twisting weak, midrib color of lower side green, upper surface smooth, lower side pubescence strong, margin undulation very weak, length 5.5, width 3.5 and petiole 1.0 cm; tip broad; base broad, flush yellowish green, young leaf anthocyanin weak.

Flower: Flowering more than once, spring and autumn, size medium.

Fruit: Size medium, skin color (mature fruit) pale green, ripened golden red, stalk cavity medium deep, fruit shape round oval, base and apex round, fruit length 4.6, breadth 4.4, stalk 0.7 cm, skin thin, surface relief medium smooth, longitudinal streaks present, prominence weak, color reddish, fruit muskiness weak.

Flesh: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor medium, pleasant with sweet and wonderful.

Seed: Stone size medium small.

Yield: High.

Leading characters: Gola Lemai plant small tree of medium size, well branched, spreading semi droopy, medium spiny, spines medium, fruit size medium, round oval, stalk cavity medium deep, pale green at maturity, reddish brown at ripening, longitudinal reddish streaks weak, surface smooth, flesh white, texture fine, uniform, stone medium small, aroma medium, flavor pleasant.

GOLA KARIRI

Origin and adaptation: Gola Kariri, a medium commercial variety of *Ber*, (*Zizyphus mauritania*); a chance seedling selection, medium early variety of Sindh.

Maintainer: Jujuba Research Station, HRI, Mirpurkhas and GPU, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, shoot thin, well branched, branch attitude spreading droopy, medium spiny, spine size medium, young shoot color brown.

Leaf: Foliage medium dense, light green, leaf size medium, midrib curvature medium, twisting weak, midrib color of lower side green, upper surface smooth, lower side pubescence strong, margin undulation very weak, length 8.5, width 5.5 and petiole 1.0 cm; tip broad; base broad, flush yellowish green, young leaf anthocyanin weak.

Flower: Flowering more than once, spring and autumn, size medium.

Fruit: Size medium, skin color (mature fruit) pale green, ripened reddish brown, stalk cavity narrow, fruit shape round oval, base and apex round, fruit length 4.0, breadth 3.2, stalk 1.0 cm, skin thin, surface relief medium smooth, longitudinal streaks present, prominence medium, color reddish, fruit muskiness weak.

Flesh: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor medium, pleasant with sweet and wonderful.

Seed: Stone size medium small.

Yield: High.

Leading characters: Gola Kariri plant small tree of medium size, well branched, spreading droopy, medium spiny, spines medium, fruit size medium, round oval, pale green at maturity, reddish brown at ripening, longitudinal reddish streaks prominence medium, surface smooth, flesh white, texture fine, uniform, stone medium small, aroma medium, flavor pleasant.

GOLA SAFINA

Origin and adaptation: Gola Safina, a medium commercial variety of *Ber*, (*Zizyphus mauritania*); a chance seedling selection, medium early variety of Sindh.

Maintainer: Jujuba Research Station, HRI, Mirpurkhas and GPU, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, shoot thin, well branched, branch attitude spreading droopy, medium spiny, spine size medium, young shoot color brown.

Leaf: Foliage medium dense, medium green; leaf size medium, midrib curvature medium, twisting weak, midrib color of lower side green, upper surface smooth, lower side pubescence strong, margin undulation very weak, length 7.5, width 5.5 and petiole 2.0 cm; tip broad; base broad, flush yellowish green, young leaf anthocyanin weak.

Flower: Flowering more than once, spring and autumn, size medium.

Fruit: Size medium, skin color (mature fruit) white, ripened whitish, stalk cavity narrow, fruit shape round oval, base and apex round, fruit length 4.2, breadth 4.0, stalk 0.7 cm, skin thin, surface relief medium smooth, longitudinal streaks present, prominence medium, color reddish, fruit muskiness weak.

Flesh: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor medium, pleasant with sweet and wonderful.

Seed: Stone size medium small.

Yield: High.

Leading characters: Gola Safina plant small tree of medium size, well branched, spreading droopy, medium spiny, spines medium, fruit size medium, round oval, stalk cavity narrow, pale green at maturity, reddish brown at ripening, longitudinal reddish streaks prominence medium, surface smooth, flesh white, texture fine, uniform, stone medium small, aroma medium, flavor pleasant.

GOLA SOGHAT

Origin and adaptation: Gola Soghat, a medium commercial variety of *Ber*, (*Zizyphus mauritania*); a chance seedling selection, medium early variety of Sindh.

Maintainer: Jujuba Research Station, HRI, Mirpurkhas and GPU, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, shoot thin, well branched, branch attitude spreading semi droopy, medium spiny, spine size medium, young shoot color brown.

Leaf: Foliage medium dense, light green; leaf size medium, midrib curvature medium, twisting weak, midrib color of lower side green, upper surface smooth, lower side pubescence strong, margin undulation very weak, length 6.4, width 4.5 and petiole 1.0 cm; tip broad; base broad, flush yellowish green, young leaf anthocyanin weak.

Flower: Flowering more than once, spring and autumn, size medium.

Fruit: Size medium, skin color (mature fruit) pale green, ripened reddish brown, stalk cavity narrow, fruit shape round oval, base and apex round, fruit length 4.0, breadth 3.5, stalk 0.5 cm, skin thin, surface relief medium smooth, longitudinal streaks very weak, color reddish, fruit muskiness weak.

Flesh: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor medium, pleasant with sweet and wonderful.

Seed: Stone size medium small.

Yield: High.

Leading characters: Gola Soghat plant small tree of medium size, well branched, spreading, semi droopy, medium spiny, spines medium, fruit size medium, round oval, stalk cavity narrow, pale green at maturity, reddish brown at ripening, longitudinal reddish streaks prominence very weak, surface smooth, flesh white, texture fine, uniform, stone medium small, aroma medium, flavor pleasant.

GOLA WHITE

Origin and adaptation: Gola White, a medium commercial variety of *Ber*, (*Zizyphus mauritania*); a chance seedling selection, medium early variety of Sindh.

Maintainer: Jujuba Research Station, HRI, Mirpurkhas and GPU, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, shoot thin, well branched, branch attitude spreading droopy, medium spiny, spine size medium, young shoot color brown.

Leaf: Foliage medium dense, dark green; leaf size medium, midrib curvature medium, twisting weak, midrib color of lower side green, upper surface smooth, lower side pubescence strong, margin undulation very weak, length 7.0, width 5.0 and petiole 1.5 cm; tip broad; base broad, flush yellowish green, young leaf anthocyanin weak.

Flower: Flowering more than once, spring and autumn, size medium.

Fruit: Size medium, skin color (mature fruit) pale green, ripened reddish brown, stalk cavity shallow, fruit roundish, broad in the middle, base and apex narrow round, fruit length 3.7, breadth 3.8, stalk 0.5 cm, skin thin, surface relief medium smooth, longitudinal streaks absent, fruit muskiness weak.

Flesh: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor medium, pleasant with sweet and pleasant.

Seed: Stone size medium small.

Yield: High.

Leading characters: Gola White plant small tree of medium size, well branched, droopy, medium spiny, spines medium, fruit size medium, round oval, pale green at maturity, reddish brown at ripening, longitudinal reddish streaks absent, surface smooth, flesh white, texture fine, uniform, stone medium small, aroma medium, flavor pleasant.

BAHAWALPUR SELECTION

Origin and adaptation: Bahawalpur Selection (Mehmoodwali), a medium commercial variety of *Ber*, (*Zizyphus mauritania*); a chance seedling selection, early season variety of Bahawalpur zone.

Maintainer: Horticultural Research Station, Bahawalpur and GPU, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, shoot thin, well branched, branch attitude spreading droopy, medium spiny, spine size medium, young shoot color brown.

Leaf: Foliage medium dense, light green, leaf size medium, midrib curvature medium, twisting weak, midrib color of lower side green, upper surface smooth, lower side pubescence strong, margin undulation very weak, length 8.5, width 5.5 and petiole 1.0 cm; tip broad; base broad, flush yellowish green, young leaf anthocyanin weak.

Flower: Flowering more than once, spring and autumn, size medium.

Fruit: Size medium bold, skin color (mature fruit) pale green, ripened reddish brown, stalk cavity narrow, fruit shape round oval, base and apex round, fruit length 4.0, breadth 3.2, stalk 1.0 cm, skin thin, surface relief medium smooth, some dark spots on skin present, fruit muskiness weak.

Flesh: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor medium, pleasant with sweet and wonderful.

Seed: Stone size medium small, rough and pitted.

Yield: High.

Leading characters: Bahawalpur selection (Mehmoodwali) plant small tree of medium size, well branched, spreading droopy, medium spiny, spines medium, fruit size medium, round oval, pale green at maturity, reddish brown at ripening, longitudinal reddish streaks prominent medium, surface smooth, flesh white, texture fine, uniform, stone medium small, aroma medium, flavor pleasant.

SUFI LOCAL

Origin and adaptation: Sufilocal, a medium commercial variety of *Ber*, (*Zizyphus mauritania*); a chance seedling selection, medium early variety of Sindh.

Maintainer: Jujuba Research Station, HRI, Mirpurkhas and GPU, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, well branched, shoot thin, branch attitude semi spreading to semi droopy, medium spiny, spine size medium fine, young shoot color brown.

Leaf: Foliage medium dense, dark green; leaf size medium large, midrib curvature medium, twisting weak, midrib color of lower side green, upper surface smooth, lower side pubescence strong, margin undulation weak, length 9.0, width 5.0 and petiole 1.5 cm; tip pointed; base acute, flush yellowish green, young leaf anthocyanin weak.

Flower: Flowering more than once, spring and autumn, size medium.

Fruit: Size medium, skin color (mature fruit) golden green, ripened golden brown, stalk cavity small, fruit shape round oval, medium broad, base and apex round, fruit length 3.7, breadth 2.8, stalk 1.0 cm, skin thin, surface relief medium smooth, longitudinal streaks present, medium prominent, color reddish, fruit muskiness weak.

Flesh: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor medium, pleasant with sweet and wonderful.

Seed: Stone size medium small.

Yield: High.

Leading characters: **Sufi Local** plant small tree of medium size, well branched, semi spreading to semi droopy, medium spiny, spines medium fine, fruit smaller than Sufi Sangar, round oval, golden green at maturity, golden brown at ripening, longitudinal reddish streaks medium prominent, surface smooth, flesh white, texture fine, uniform, stone size medium small, aroma medium, flavor pleasant.

SUFI SANGAR

Origin and adaptation: Sufi Sangar, a medium commercial variety of *Ber*, (*Zizyphus mauritania*); a chance seedling selection, medium early variety of Sindh.

Maintainer: Jujuba Research Station, HRI, Mirpurkhas and GPU, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, well branched, shoot thin, semi spreading to semi droopy, medium spiny, spine size medium fine, young shoot color dark brown.

Leaf: Foliage medium dense, medium green; leaf size medium, midrib curvature medium, twisting medium, midrib color of lower side green, upper surface smooth, lower side pubescence very strong, margin undulation weak, length 7.0, width 4.0 and petiole 1.5 cm; tip roundish; base obtuse, flush yellowish green, young leaf anthocyanin weak.

Flower: Flowering more than once, spring and autumn, size medium.

Fruit: Size medium bold, skin color (mature fruit) pale green, ripened reddish brown, stalk cavity small, fruit shape round oval, medium broad, base and apex round, fruit length 5.2, breadth 3.8, stalk 1.5 cm, skin thin, surface relief medium smooth, longitudinal streaks present, medium prominent, color reddish, fruit muskiness weak.

Flesh: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak to medium, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor medium, pleasant with sweet and wonderful.

Seed: Stone size medium small.

Yield: High.

Leading characters: Sufi Sangar plant small tree of medium size, well branched, semi spreading to semi droopy, medium spiny, spines medium fine, fruit medium bold, round oval, medium broad, pale green at maturity, red at ripening, longitudinal reddish streaks medium prominent, surface smooth, flesh white, texture fine, uniform, puffiness weak to medium, stone size medium small, aroma medium, flavor pleasant.

SUFI SIALKOTI

Origin and adaptation: Sufi Sialkoti, a medium commercial variety of *Ber*, (*Zizyphus mauritania*); a chance seedling selection, medium early variety of Sindh.

Maintainer: Jujuba Research Station, HRI, Mirpurkhas and GPU, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, well branched, shoot thin, well branched, branch attitude semi spreading droopy, medium spiny, spine size medium fine, young shoot color brown.

Leaf: Foliage medium dense, dark green; leaf size medium, midrib curvature medium, twisting weak, midrib color of lower side green, upper surface smooth, lower side pubescence strong, margin undulation very weak, length 6.0, width 3.5 and petiole 2.0 cm; tip broad; base broad, flush yellowish green, young leaf anthocyanin weak.

Flower: Flowering more than once, spring and autumn, size medium.

Fruit: Size medium, skin color (mature fruit) pale green, ripened reddish brown, stalk cavity wide, fruit shape round oval, base and apex round, fruit length 4.0, breadth 3.2, stalk 1.0 cm, skin thin, surface relief medium smooth, longitudinal streaks present, prominence medium, color reddish, fruit muskiness weak.

Flesh: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor medium, pleasant with sweet and wonderful.

Seed: Stone size medium small.

Yield: High.

Leading characters: Sufi Sialkoti plant small tree of medium size, well branched, droopy, medium spiny, spines medium fine, fruit size medium, round oval, pale green at maturity, reddish brown at ripening, longitudinal reddish streaks prominence medium, surface smooth, flesh white, texture fine, uniform, stone medium small, aroma medium, flavor pleasant.

UMRAN

Origin and adaptation: Umarn, a medium commercial variety of *Ber*, (*Zizyphus mauritania*); a chance seedling selection, medium late variety of Sindh and Punjab.

Maintainer: Jujube Research Station, (HRI), Tandojam, HRS, Bahawalpur and GPU, DI Khan.

Tree: Small tree, medium height, vigorous growing, stem medium thin, well branched, shoot thin, branch attitude semi spreading droopy, sparsely spiny, spine size medium fine, young shoot color brown.

Leaf: Foliage medium dense, dark green; leaf size medium small, midrib curvature medium, twisting weak, midrib color of lower side green, upper surface smooth, lower side pubescence strong, margin undulation weak, length 6.0, width 4.0 and petiole 1.0 cm; tip pointed; base acute, flush yellowish green, young leaf anthocyanin weak.

Flower: Flowering more than once, spring and autumn, size medium.

Fruit: Size medium, skin color (mature fruit) pale green, on ripening golden yellow turning chocolate brown, stalk cavity small, fruit shape round oval, medium broad, base and apex round, fruit length 3.7, breadth 2.8, stalk 0.8 cm, skin thin, surface relief medium smooth, longitudinal streaks present, prominence strong, color reddish, fruit muskiness weak.

Flesh: Flesh color white, color uniform, discoloration after cutting very mild, outer flesh grittiness very weak, puffiness weak, juiciness medium, acidity low, sweetness high, texture fine, medium firm, flavor medium, pleasant with sweet and wonderful.

Seed: Stone size medium small.

Yield: High.

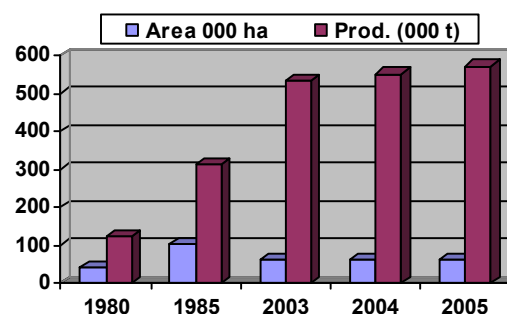
Leading characters: **Umran** plant small tree of medium size, well branched, droopy, sparsely spiny, spines medium fine, fruit bigger than Sufilocal, round oval, pale green at maturity, turning golden yellow turning chocolate brown at ripening, longitudinal reddish streaks prominence strong, surface smooth, flesh white, texture fine, uniform, stone size medium small, aroma medium, flavor pleasant.

GUAVA PRODUCTION AND PROPAGATION

AMROOD

The guava (*Psidium guajava*) is a very popular common fruit of the Indo-Pak sub-continent and is grown almost universally in all tropical, sub-tropical and Mediterranean

regions. In Pakistan it is gaining importance as the area under guava is increasing gradually and has attained 4th position after citrus, mango and date palm. It covered 17,300 ha (43, 200 acres) in 1980-81 and has jumped to about 63,000 ha (157,000 acres) in 2002-03. Similarly the production has also registered a significant increase from 123,600 tons in 1980-81 to 531,600 tons in 2002-03.



The name Guava comes from the common name of the fruit in the Spanish speaking tropical America the *Guayaba*. The Guava (*Psidium guajava* L) belongs to the myrtle family (*Myrtaceae*). The place of origin is uncertain but believed to be Mexico and Central America. Its cultivation started in mid 18th century and was introduced in Asia and the Indo-Pak Sub-continent during late 18th century. It is best adapted to warm and tropical climates under both humid and dry climates and can successfully tolerate short spells of frost. The tree can withstand temporary water

logging and very high temperature.

The guava is evergreen small vigorous tree of medium height, with medium to shallow-root system, semi-spreading to spreading branches, smooth, thin, copper-colored bark that peels off in thin flakes, showing the greenish layer beneath giving attractive, bright “bony” looks. Young shoots start arising close to the ground and are quadrangular and downy.

The dull-green leaves on the young twigs are opposite, short-petioled, oval to oblong-elliptic in shape, have pronounced and conspicuous parallel veins and slightly downy, finely pubescent on the underside and nearly glabrous above. The leaves emits aroma on crushing.

Faintly fragrant, the white flowers, borne singly or in small clusters in the leaf axils have 4 or 5 white petals which are quickly shed. The guavas can bloom throughout the year but the heaviest bloom occurs in spring. The tree is primarily self-fruitful although cross-pollination occurs mainly through honey bees (*Apis mellifera*) and the amount of cross-pollination ranges from 25.7 to 41.3%.

The fruit may be round, ovoid, or pear-shaped, with 4 or 5 protruding sepal remnants at the apex. The skin is thin, light-yellow oftenly blushed with pink. The flesh may be white, yellowish, light- or dark-pink, or near-red, juicy, acid, sub-acid, or sweet and flavorful. The central pulp may be slightly darker in tone, contains soft to very hard, yellowish seeds which in good varieties are fully edible. Actual seed counts have been reported to range from 112 to 535 but some guavas are seedless or nearly so. The fruit emits sweet musky odor which on over ripening becomes pungent and penetrating. The immature green fruits may be hard, gummy and sometimes very astringent. In warm weather the fruit ripens in three

months after opening of the flowers and two crops a year can be borne.

Climate and Soil

The guava can be successfully grown in both humid and dry climates. It is cold sensitive and can't resist long periods of frost. A mild tropical climate induces fruiting throughout the year while under subtropical conditions two distinct crops are obtained. Heavy rains and high humidity during ripening stage cause damage to the skin of the fruit. It is commercially grown on large areas in central Punjab, Larkana, Dadu, Mirpurkhas, Hyderabad, Bannu, Kohat and DI Khan. The guavas can be produced on a wide variety of soils doing equally well on heavy clay, clayey loam, light sandy, somewhat salty and coastal areas. The plant can tolerate soils with a pH ranging from 4.5 to 9.4. Under favorable climatic conditions the guava trees grow rapidly and bore fruit in 2 to 4 years from seed. They live 30 to 40 years but productivity may declines after the 15th year.

Propagation

The guava can be propagated both by seed as well as vegetative means by inarching, soft wood cuttings, layering, approach grafting, budding, root-suckers, root-cuttings. Budding had not been very successful.

Tree ripe fruits should be selected for seed extraction. The seed should be cleanly washed, dried in shade, packed in vapor proof container and stored in cool dry place. The seeds remain viable for many months. They take about 2 to 3 weeks to germinate. A brief pretreatment with sulfuric acid, or boiling for 5 minutes, or soaking for 2 weeks, will hasten

germination. The seedlings, when 10-15 cm tall are transplanted in nursery rows or in pots for grafting. The nursery plants are set out in the field when 1 or 2 years old. Although majority of guava nursery plants are raised from seed but the trees cannot be depended upon to come true from seed because of occurring considerable amount of cross-pollination. For getting true-to-type trees, vegetative propagation is practiced particularly in Hyderabad and Larkana areas.

The tree can be grown from root cuttings. Pieces of any roots except the smallest and the very large, cut into 12-20 cm lengths, are placed flat in a prepared bed and covered with 5-10 cm of soil which must be kept moist. Another method of guava propagation is using root-suckers. The lateral roots of a good tree are exposed in the ground 0.5-1.0 m away from the tree trunk. Cut wounds are made on them and left to sprout. The cut ends will sprout and can be severed from the mother tree and dug up and transplanted.

Inarching or approach grafting at the start of monsoon is a good method of propagation. Vigorous seedlings 1.5-2.5 cm thick are used as rootstocks. The scion should not be separated from the mother tree for at least three months. When the scion is separated from the mother plant, the graft is placed under shade for some days to harden before planting it out in the field.

Ground layering is common method of vegetative propagation in Larkana and Hyderabad. A branch of the tree is bent to the ground and buried leaving out the growing tip. Sometimes a slanting cut is made on the covered portion to induce rooting.

Pruned branches and soft wood cuttings may serve as propagating

materials. Cuttings of half-ripened wood will root with bottom heat or rooting-hormone treatment. Under monsoon conditions shoot cuttings 45-60 cm long, stuck into 30-cm long black plastic bags filled with soil, readily roots without chemical treatment.

The trees grown from cuttings or air- and ground-layering have one big disadvantage that they have no taproot but due to shallow root system are apt to be blown down in the first 2 or 3 years. For this reason, budding and grafting are preferred.

Cultural Practices

The guava trees are frequently planted too close but the optimum distance between the trees should not be less than 8 m. However guavas can be planted closer when planted as fillers in mango or other orchards and as hedge trees. The orientation of orchard rows should always run north and south so that each tree receives the maximum sunlight.

The guava bears fruit on current season's growth. Light pruning is always recommended to develop a strong framework. Suckers which sprout all the year round particularly during spring and monsoon should also be eliminated around the base. In Palestine, the trees are cut back to 6 ½ ft (2 m) every other spring to facilitate harvesting without ladders. Fruits are borne by new shoots from mature wood. If trees bear too heavily, the branches may break. Therefore, thinning is recommended and results in larger fruits. The guava trees older than 15 years should be pruned heavily to rejuvenate them for increasing fruit yield.

The guava is a very hardy tree and drought-tolerant but must receive irrigation once in every 10-15 days for the first three to four years. In dry

regions lack of irrigation during the period of fruit development will cause the fruits to be deficient in size. In areas receiving low rainfall, the guava will require additional irrigation at an interval of 15-20 days in summer and one each month in winter.

To increase yield and quality the guava trees respond to a complete fertilizer mix applied once a month during the first year and every other month the second year after transplanting. The dose may be 300 to 700 g per tree depending on age of the tree. Nutritional sprays providing Nitrogen, copper, potassium and zinc also produce good results.

During the first three to four years when the guava plants have not grown to size, an intercrop preferably of vegetables or pulses can be planted to increase the farm income. The plants should be kept weed free. Moderate opening of the soil following all irrigations specifically during fruit development and maturity keeps the weeds under control and is said to help improve the size and quality of the fruit.

Cropping and Yield

Like ber the guava is also the earliest fruit plant to bear. Sometimes the seedling plants flower even in the nursery. Generally, there are 2 crops per year in Punjab and NWFP; the main crop ripens in mid-winter and the fruits are of the best quality. A second crop is home in the rainy season but the fruits usually are of poor quality due to heavy attack of the fruit fly. Growers usually withhold irrigation after December or January or root-prune the trees in order to avoid a second crop. The trees will shed many leaves and any fruits set will drop. Full grown trees during winter crop may bear 300 to 500 fruits per tree whereas 100-300 fruits in the rainy season. The yields of course vary with the cultivar and cultural treatment.

Handling and Marketing

Ripe guavas bruise easily and are highly perishable. For fresh marketing and transporting to other cities, the fruits must be hand picked preferably clipped when full grown but under-ripe, and handled with great care. After grading for size, the fruits should be wrapped individually in tissue and packed in suitable containers (paper or wooden crates) with padding on each side. Fruits picked when yellow-green will ripen in 4-5 days during transit to long distance markets. In the kitchen placing a ripe banana or apple in the basket or crate and covered with paper will hasten ripening. Fruits for processing can be harvested by mechanically shaking the tree and collecting the fruit in plastic nets.

Pests and Diseases

In Pakistan fruit fly is the most serious and destructive insect to guavas. Ripe fruits particularly from summer crop will be found infested with the larvae and totally unusable except as feed for sheep and goats. Failure to control the insect has forced guava growers to stop taking a summer crop. They concentrate on getting a good winter crop. The female insect punctures the fruit to lay eggs. The larvae hatch after a few days and start eating the flesh of the fruit. Such fruits drop and quickly rot on the ground. To avoid fruit fly damage, all fallen fruit should be collected and buried in the ground and the trees should be sprayed at least twice with suitable insecticides during the early days of fruit set. In Dade County, Florida, wasps have been employed to attack the larvae and pupae of the fruit fly and this year have somewhat reduced the menace. Fruits punctured by insects are subject to rot (caused by the fungus).

The other problem noticed recently is the dieback of guava plants.

Previously this phenomenon was noticed on *shisham* (*Dilbergia*). The plants start dying slowly. The Dieback is more pronounced where spring and early summer irrigation is stopped to discourage the summer crop in the hope of getting a good winter crop. It is supposed that due to drought stress the leaves fall. In the absence of foliage and moisture stress the tree can not transpire sufficiently and severely heat stressed. The high internal temperature affects the metabolic system of the tree adversely and consequently they start dying.

Severe losses to mature fruit have also been noticed caused by bird attack particularly the bulbul, parrot and sparrows. The control measure include the efforts made to protect the crop by nets or noisemakers.

Uses

Fruit: In Pakistan raw guavas are eaten out-of-hand. They are in very high demand particularly during fasting month (*Ramzan*) for making fruit chat for *aftari* in the evening. Their use in making squashes and nectar is very limited but increasing. In many other countries guavas are preferred seeded and served sliced as dessert or in salads. Sometimes the fruit is cooked and cooking eliminates the strong odor. A standard dessert throughout Latin America and the Spanish-speaking islands of the West Indies is stewed guava shells (cascos de guayaba), that is, guava halves with the central seed pulp removed, strained and added to the shells while cooking to enrich the syrup. The canned product is widely sold and the shells can also be quick-frozen. They are often served with cream cheese. Sometimes guavas are canned whole or cut in half without seed removal.

Bars of thick, rich guava paste and guava cheese are staple sweets, and guava jelly is almost universally marketed. Guava juice, made by

boiling sliced, unseeded guavas and straining, is much used in Hawaii in punch and ice cream sodas. A clear guava juice with all the ascorbic acid and other properties undamaged by excessive heat, is made in South Africa by trimming and mincing guavas, mixing with a natural fungal enzyme (now available under various trade names), letting stand for 18 hours at 120° to 130° F (49°-54° C) and filtering. It is made into sirup for use on waffles, ice cream, puddings and in milkshakes. Guava juice and nectar are among the numerous popular canned or bottled fruit beverages of the Caribbean area. After washing and trimming of the floral remnants, whole guavas in sirup or merely sprinkled with sugar can be put into plastic bags and quick-frozen.

There are innumerable recipes for utilizing guavas in pies, cakes, puddings, sauce, ice cream, jam, butter, marmalade, chutney, relish, catsup, and other products. In India, discoloration in canned guavas has been overcome by adding 0.06% citric acid and 0.125% ascorbic acid to the sirup. For pink sherbet, French researchers recommend 2 parts of the cultivar 'Acid Speer' and 6 parts 'Stone'. For white or pale-yellow sherbet, 2 parts 'Supreme' and 4 parts 'Large White'. In South Africa, a baby-food manufacturer markets a guava-tapioca product, and a guava extract prepared from small and overripe fruits is used as an ascorbic-acid enrichment for soft drinks and various foods.

Dehydrated guavas may be reduced to a powder which can be used to flavor ice cream, confections and fruit juices, or boiled with sugar to make jelly, or utilized as pectin to make jelly of low-pectin fruits. India finds it practical to dehydrate guavas during the seasonal glut for jelly-manufacture in the off-season. In 1947, Hawaii began sea shipment of

frozen guava juice and puree in 5-gallon cans to processors on the mainland of the United States. Since 1975, Brazil has been exporting large quantities of guava paste, concentrated guava pulp, and guava shells not only to the United States but to Europe, the Middle East, Africa and Japan.

Canned, frozen guava nectar is an important product in Hawaii and Puerto Rico but may be excessively gritty unless stone cells from the outer flesh and skin are reduced by use of a stone mill or removed by centrifuging. Green mature guavas can be utilized as a source of pectin, yielding somewhat more and higher quality pectin than ripe fruits.

Ascorbic acid—mainly in the skin, secondly in the firm flesh, and little in the central pulp—varies from 56 to 600 mg. It may range up to 350-450 mg in nearly ripe fruit. When specimens of the same lot of fruits are fully ripe and soft, it may decline to 50-100 mg. Canning or other heat processing destroys about 50% of the ascorbic acid. Guava powder containing 2,500-3,000 mg ascorbic acid was commonly added to military rations in World War II. Guava seeds contain 14% of an aromatic oil, 15% protein and 13% starch. The strong odor of the fruit is attributed to carbonyl compounds.

Wood: The wood is yellow to reddish, fine-grained, compact, moderately strong, weighs 650-750 kg per cubic meter; is durable indoors; used in carpentry and turnery. Though it may warp on seasoning, it is much in demand in Malaya for handles; in India, it is valued for engravings. Guatemalans use guava wood to make spinning tops, and in El Salvador it is fashioned into hair combs which are perishable when wet. It is good fuelwood. and also a source of charcoal.

Leaves and bark: The leaves and bark are rich in tannin (10% in the

leaves on a dry weight basis, 11-30% in the bark). The bark is used in Central America for tanning hides. Malaysians use the leaves with other plant materials to make a black dye for silk. In South East Asia, the leaves are employed to give a black color to cotton; and in Indonesia, they serve to dye matting.

Medicinal Uses: The roots, bark, leaves and immature fruits, because of their astringency, are commonly employed to halt gastroenteritis, diarrhea and dysentery, throughout the tropics. Crushed leaves are applied on wounds, ulcers and rheumatic places, and leaves are chewed to relieve toothache. The leaf decoction is taken as a remedy for coughs, throat and chest ailments, gargled to relieve oral ulcers and inflamed gums; and also taken as an emmenagogue and vermifuge, and treatment for leucorrhea. It has been effective in halting vomiting and diarrhea in cholera patients. It is also applied on skin diseases. A decoction of the new shoots is taken as a febrifuge. The leaf infusion is prescribed in India in cerebral ailments, nephritis and cachexia. An extract is given in epilepsy and chorea and a tincture is rubbed on the spine of children in convulsions. A combined decoction of leaves and bark is given to expel the placenta after childbirth.

BER PRODUCTION AND PROPAGATION

The Ber (*Ziziphus mauritiana*) is a fast growing indigenous tree having its origin in Indo-Pakistan Subcontinent, and China. The tree occurs growing in several countries like Pakistan, China, India, Syria, Malacca, Australia, Malayan archipelago and Mediterranean regions. It has also been found growing in many parts of Afghanistan, Iran and sub-tropical regions of Russia, Uzbekistan and Azerbaijan. In China it is called Chinese date. The Ber in Indo-Pakistan Subcontinent has been in use for almost 4000 years.

The ber tree or Indian jujube is a small spreading tree usually with vine like and drooping branches. The tree is 7 – 12 m in height and the trunk, 30 cm in diameter. Branches are slender and downy, bearing paired, brown spines, one straight and the other slightly hooked. It has a wide spreading crown. The leaves are green to dark green, and densely tomentose (hairy) on the under-side. The leaves are shed during summer or after crop harvesting. The flowers are borne in autumn in clusters or rarely solitary. The flowers are pollinated by honey bees (*Apis* spp.) and house flies (*Musca domestica*). The fruit ber is a drupe ripening in winter or early spring. It is generally slightly juicy with two or three seeds in a hard endocarp. The fruits are variable in shape, colour and size. Round, oval or oblong in shape, the fruits may be yellow, green, reddish or purple/dark brown in colour. Wild fruits are 1.5 - 2.5 cm in length and 1-2 cm in diameter; however improved cultivars can be up to 5 cm in length. The ripe fruit (drupe) is filled with a juicy, hard or soft, sweet-tasting pulp.

Propagation

Desi or wild ber trees can be easily grown from seeds. However improved commercial ber orchards are raised by shield or patch budding. For budding, rootstock seedlings are raised by sowing seeds collected from wild or desi trees. Germination of the seeds usually occurs within 1 - 4 weeks. Seed can retain its viability for 1 - 2.5 years provided it is kept dry. Seeds should be planted approximately 0.6 - 2.0 cm deep in containers filled with soil-manure mixture. Seeds should be extracted from fruits taken from vigorous trees. Budding is carried out on 100 day-old rootstock seedlings during the summer. Scion budsticks are taken from the trees of identified desired varieties having a record of good production. The budded trees can be planted out in the field about 30 days after budding in the nursery. Alternatively, 100 day-old rootstock seedlings can be planted out in the field and budded in situ during the following summer. When planted out, the young trees require adequate soil moisture to become well established, but mature trees can withstand drought quite well

Soil and Fertilizer requirements

The tree is drought and saline tolerant and can grow on poor and degraded land but grows well on light soils. Generally the farmers do not apply any fertilizer to the Ber trees but for a good yield appropriate application of fertilizers as necessary. Regular application of FYM and chemical fertilizers is recommended. Deficiencies of N P K reduce shoot length, leaf number and size, axillary branch number, flowering and fruit set and consequently fruit yield of the tree. Under ordinary growth conditions the 100g, 100g, 150g NPK per tree may be added for a good

harvest.

Water management

Availability of irrigation water is the single most important factor determining productivity and quality of ber fruits, particularly in barani and arid areas. The appropriate application of water in rainfed as well as irrigated areas is of great importance for achieving sustained fruit production.

Weed control

Weeds are a major problem especially when they suck up the limited available moisture under barani or rainfed conditions. Effective weed control can be obtained through a combination of mechanical and manual control measures. Ploughing the inter-row spaces immediately after summer pruning, not only controls the weeds but also exposes the soil to solar radiation and kills hibernating insect pests. It also allows maximum absorption of rain-water into the soil. Ploughing must be carried out just after the weeds emerge. These operations are combined with manual weeding of the area around the trees. Herbicides can also be used to control weeds. Chemical weed control around the trees can be an effective alternative. The herbicide use must be carried out according to the recommendation of the local area experts.

Cover crops and intercrops

Although ber is a quick growing tree yet it takes four to five years after planting to cover the inter-row spaces. Moreover during this period the production also remains low hence in the mean time, considerable losses occur from these vacant interspaces. In order to

minimize the losses and to generate additional income, the farmers usually grow intercrops in the orchards. The intercrop must not be exhaustive, tall growing or insect attractant. The most suitable intercrops can be vine vegetables or edible legumes like mung beans, moth beans, cowpeas, cluster beans and oilseed crop like sesame can be grown under rainfed conditions during the summer. Chillies, other spices or medicinal herbs can be more profitable than the ordinary pulse crops or vegetables. The practice of growing intercrop in the ber orchards should be stopped as soon as the trees reach full brown stage and cover up the inter-spaces.

The fruits mature in winter to late spring. All the fruits do not ripen at the same time and should be picked 3 to 5 times in the season. Picking should be carried out selectively at the correct stage of maturity as under-ripe fruits have an acrid flavour. Ripe fruits, have a short storage life and should be stored in a cool dry place.

The Ber fruit, has a high sugar content and a high level of vitamins A & C, carotene, phosphorus and calcium. The leaves contain 6 % digestible crude protein, which is an excellent source of ascorbic acid and carotenoids.

uses

The fruit is the most well known. The fresh fruit has a mild sub-acid flavour and crisp firm flesh, it can also be boiled, stewed or baked. Other culinary uses include preparation of pickles, jams, candied fruits, beverages, ber butter and cheese-like pastes. The leaves are used as forage for cattle/sheep/goats and are also palatable for human consumption, used as a vegetable in couscous. The timber, though very hard, can be worked to make fine grained tools, boats, charcoal and poles for house building.

Roots, bark, leaves, wood, seeds and fruits are reputed to have medicinal properties. The tree is also used as a source of tannins, dyes, silk (via silkworm fodder), shellac and nectar. The ber tree can be planted in areas for soil conservation in sandy areas, as a windbreak and as living fences for stock control.

DISEASES

Powdery mildew

Powdery mildew (*Oidium erysiphoides* f. sp. *ziziphi*) is one of the most serious diseases which can cause severe losses in *ber* orchards. The intensity of such losses increases under humid conditions. The disease occurs with increased virulence during high rainfall years. The symptoms of the disease are noticed on flowers and newly set fruits. The developing young leaves show a whitish powdery mass, which causes them to shrink and defoliate. The disease also appears in the form of white powdery spots on the surface of the fruits and later covers the whole fruit surface. The spots turn into light brown to dark brown in severe cases. The infected area becomes slightly raised and rough. The affected fruits either drop off prematurely or become corky, cracked, mis-shapen and remain underdeveloped. The disease can be controlled by spraying suitable fungicides.

***Alternaria* leaf spot**

Several species of *Alternaria* cause this disease, produces small,

irregular brown spots on the upper surface of the leaves and dark brown to black spots on the lower surface. The spots coalesce forming large blighted patches and the affected leaves drop. The disease can be controlled by spraying mancozeb or other appropriate fungicide.

Insect Pests

Several insects attack *ber* trees, but only fruitfly, fruit borer, bark eating caterpillar, chafer beetle and hairy caterpillar cause significant damage.

Fruitfly

The fruitfly *Carpomyia vesuviana*, *Dacus correctus* and *D. dorsalis* (Diptera Tephritidae) infest *ber* fruits and cause very severe losses. Infestation of fruitfly starts with the onset of fruit setting. The adult female lays eggs singly by inserting its ovipositor in the developing fruit. After 2 to 5 days, the larva hatches out and starts feeding on the pulp. To prevent infestation, prophylactic sprays should be carried out. Almost all commercial cultivars are attacked by fruitfly but the extent of damage varies in different locations and years depending upon agroclimate and growing conditions.

Lac insect

Ber is a host of the lac insects, (*Kerria lacca*) and (*K. sindica*). The small insect becomes active in summer (April-May). They secrete a thick, resinous substance which envelopes their bodies. The secretions form a hard crust on the twigs, which is collected to form a commercial resin, lac. but the insect devitalises the tree and causes great loss in fruit production. It sucks the sap from the branches, and ultimately kills them. To control

this pest, the infected twigs should be cut off at the time of annual pruning and destroyed. After pruning, the trees should be sprayed with appropriate pesticide.
